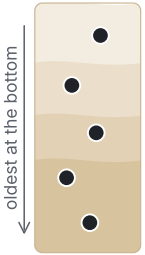
The dates each site contributes can change the story a regional curve tells

❶ One site, several dates



five dates

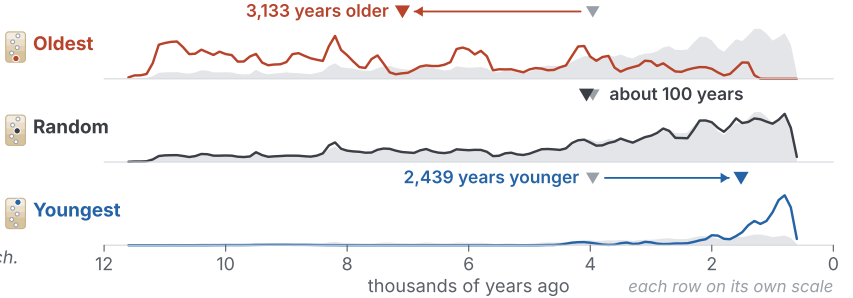
What reaches an analysis ...

- Oldest
- Youngest
- One at random
- Both extremes

and nothing in a date says which.

❷ Real dates: 540 from 68 Australian sites

if each site kept only one date (grey: all dates; ▼ average age)



❸ Simulated archives with a known history

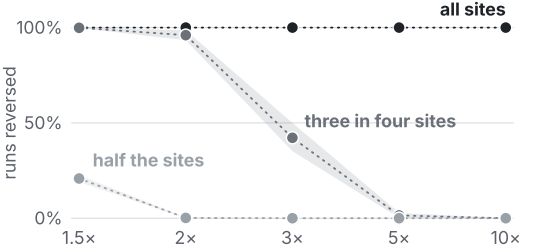
rule applied at every site; tenfold change; cells: trend concluded

	true history: Falling	Steady	Rising
Oldest *always falling*	100%	100% false trend	100% wrong direction
Youngest *always rising*	100% wrong direction	100% false trend	100%
One at random	100%	82% 18% false trend	100%
Both extremes	100%	99.9%	100%

Both extremes avoided reversals, but with gentle (1.5×) trends found the true direction in only 11–17% of runs (two random dates per site: 61–67%).

❹ Weaker trends need fewer sites to flip

affected sites keep only the extreme opposing the trend (oldest if rising, youngest if falling); the other sites keep all their dates



Many runs that escaped reversal still read the trend as steady.